Andy's Package

Quick start and notation guide

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Start a document

Copy the master Overleaf project. Its root contains `main.tex` and the three style files. Edit `main.tex`; there is no need to import the packages into every new project.

Select one notation line and one style line in the preamble. A percent sign comments out a line; move it so that exactly one style line is active. The default is a portable Times-family face, purple accents, and alternating running headers on facing pages. Overleaf's pdfLaTeX compiler works for this default.

```
\usepackage{andyspackage}
% \usepackage[kripke]{andyspackage}
% \usepackage[kripke,prooftrees]{andyspackage}

\usepackage[header=book,font=times]{andysnotes}
% \usepackage[header=plain,font=times]{andysnotes}
% \usepackage[header=minimal,font=times]{andysnotes}
% \usepackage[header=none,font=times]{andysnotes}
% \usepackage[header=plain,font=typewriter,accent=black,
%   sectionrules=false]{andysnotes}
% \usepackage{andyspics}
```

The `book` header alternates section and title text above a double rule; `plain` uses a simple running header; `minimal` shows title, author, class, date, and the current section index; `none` removes the running header. All four styles keep the page number in the footer. Add `studentid=true` to the package options and set `\AndyStudentID{...}` to show a student ID in the minimal header. The first content page is numbered 0 by the starter file. Change `\setcounter{page}{0}` to 1 if a conventional first page suits the project.

Choose only the title fields you need

A title and author suffice. The subtitle, course, assignment, due date, and affiliation are independent optional commands. Use `\AndyMakeTitle` for a title above the first section, or `\AndyMakeTitlePage` for a separate cover. Uncomment `\tableofcontents` only when useful.

```
\title{A Working Title}
\author{Andrew M. Hagy}
\date{}
% \AndyCourse{PHIL 4770}
% \AndyDue{Friday, October 2, 2026}
\begin{document}
```

```
\setcounter{page}{0}
\AndyMakeTitle
% \tableofcontents
\section{Introduction}
\subsection{First idea}
\end{document}
```

Section headings are unnumbered on the page, but they advance the section counter. Thus the first subsection prints **1.1**, and the first definition there prints **1.1.1**. A definition, lemma, proposition, claim, example, remark, note, exercise, chaff, or aside takes the next number in this shared sequence. Theorems and conjectures have separate global sequences. A corollary after Theorem 1 prints **Corollary 1.i**.

Write with the notation

$$\Gamma \vdash \varphi, \qquad \Gamma \vDash \varphi, \qquad \mathscr{M} \models \varphi, \qquad w \Vdash \varphi, \qquad \mathbb{R} \subseteq \mathbb{C}.$$

The source of that line uses the meaningful names `\proves`, `\entails`, `\satisfies`, `\forces`, `\RR`, and `\CC`. The full source-to-symbol-to-standard-LaTeX map is in `docs/cheatsheet.md`. Times remains the default font. Select `font=typewriter` for the alternate inspired by typewritten mathematics dissertations: monospaced prose with traditional mathematical symbols and more generous line spacing.

Use `\begin{DefinitionBlock}` to highlight a statement; the unboxed `definition` advances the same counter. A `ProofBlock` uses thin separator lines inside a theorem rather than a second full frame. This lets long proofs cross a page without being moved intact to the next sheet. The abbreviations WLOG, WTS, WRT, and SFSOC retain their small-cap styling.

Choose proof marks and provenance

Ordinary proofs end with a striped tombstone labelled `QED`. A hollow tombstone means quoted from or closely following a source, a striped tombstone means a general proof, and a solid purple tombstone marks a highly original contribution. The tag below it records provenance: `QED` is human proof, `XM` is *Ex machina*, and `AXM` is *Auxilium ex machina*.

```
\AndyBookMark[QED]
\AndyGeneralMark[AXM]
\AndyOriginalMark[QED]
\AndyTombstoneFootnote
```

For the two directions of a biconditional, use:

```
\begin{IffForwardProof}[QED]
Assume $P$ and prove $Q$.
\end{IffForwardProof}
\begin{IffBackwardProof}[AXM]
Assume $Q$ and prove $P$.
\end{IffBackwardProof}
```

Each environment ends with a half-striped tombstone and a directional arrow.

For casework, give the number of cases once. The environment prints “Proceed by cases.” and numbers cases with lowercase Roman numerals. The tombstone fill advances from left to right.

```
\begin{CaseProof}{3}
  \begin{AndyCase}First case.\end{AndyCase}
  \begin{AndyCase}Second case.\end{AndyCase}
  \begin{AndyCase}Third case.\end{AndyCase}
\end{CaseProof}
```

Use `\AndyProofNotationAppendix` for the long explanatory note and `\AndyOrnamentBreak` for the optional Angus-inspired divider. The living conventions and image-inclusion example are in `docs/ETIQUETTE.md`.

Prompt for transcribing handwritten notes

Copy the text below into a chat when asking an LLM to typeset your notes. The editable source is `docs/llm-prompt.txt`; the cheat sheet points back to the same file.

```
Typeset my handwritten mathematics and logic notes as a LaTeX handout for
Overleaf. This project contains andyspackage.sty and andysnotes.sty. Load
andyspackage before andysnotes; enable the [kripke] option only for Kripke
helpers and [prooftrees] only for bussproofs derivations. Load andyspics only
if a figure, diagram, or plot needs it.

Use commands actually defined in andyspackage wherever they fit. For
semantic consequence write \Gamma\entails\varphi; for truth in a structure
write \Str\satisfies\varphi; for syntactic derivability write
\Gamma\proves\varphi; for Kripke forcing write w\forces\varphi. Use
\NN, \ZZ, \QQ, \RR, \CC for number systems, and \tuple, \struc, \set
when appropriate. Prefer these aliases to bare \vDash, \models, \vdash,
or \mathbb{R}; use ordinary LaTeX if no custom command exists. Never
invent a package command. Consult docs/cheatsheet.md for the command list.

Preserve the order and mathematical claims in the notes. Mark illegible
writing in source comments beginning "% UNCLEAR:" and list ambiguities
after the complete main.tex. Do not silently repair a proof or change a
definition. Use an ordinary proof environment for long proofs, or
ProofBlock for a proof nested in a boxed statement so pages can break.
Return the complete main.tex and a brief list of required project files.
```

Citations and bibliography

Footnotes are useful for comments on the text and can contain a citation. For a short handout, keep references in `main.tex` with `\bibitem`. For an Angus-inspired alphabetic bibliography, turn on the optional bibliography setting in `andysnotes` and add a `references.bib` file; see `docs/bibliography-biblatex.tex`. The same `\cite{key}` works in both cases. A journal or instructor's required reference style takes precedence.

Definitions of symbols

This appendix distinguishes the mathematical relations named in the package. The descriptions are conventions for these notes, not claims that every author uses the same symbols.

$\in$ Membership: $x \in A$ says that x is an element of A.

$\subseteq$ Inclusion: $A \subseteq B$ allows $A = B$.

$\subsetneq$ Proper inclusion: $A \subsetneq B$ requires $A \subseteq B$ and $A \neq B$. The bare symbol $\subset$ is used differently by different authors; prefer $\subsetneq$ when strictness matters.

$\vdash$ Syntactic derivability: $\Gamma \vdash \varphi$ states that a proof of φ from Γ exists in the chosen deductive system.

$\vDash$ Semantic consequence: $\Gamma \vDash \varphi$ states that every relevant structure satisfying Γ also satisfies φ.

$\models$ Satisfaction: $\mathscr{M} \models \varphi$ says that a particular structure $\mathscr{M}$ satisfies φ. The glyph closely resembles or coincides with the previous one; the custom command records which relation is intended.

$\Vdash$ Forcing: $w \Vdash \varphi$ uses the forcing relation appropriate to the stated Kripke or poset semantics.

$\succeq$ Accessibility in the chosen Kripke orientation, written `\kripkesucceq`. The legacy `\Acc` is available with the `[kripke]` option. Peano successor remains S and is not assigned this relation symbol.

$\geqslant$ Slanted greater-than-or-equal, written `\gecurly`. This is an order glyph, not the Kripke relation.

$\langle A, R \rangle$
A displayed structure or tuple. `\struc{A,R}` and `\tuple{A,R}` print the same brackets while recording the intended role in source.

A.1 A two-item check

Decide whether each statement is true or false:

$$(1)\ \emptyset \in \emptyset, \qquad (2)\ \emptyset \subset \emptyset.$$

The first is false: the empty set has no elements. The second depends on the author's convention for $\subset$: it is true if equality is allowed and false if strict inclusion is intended. Written without ambiguity, $\emptyset \subseteq \emptyset$ is true and $\emptyset \subsetneq \emptyset$ is false.